

Robot Gesture Control — Command Table

Version: 2025-10-09 • Owner: Team Orangeeeeeeeee

Notes: Sharp turn is one-shot (edge-triggered). Smooth turn is continuous and I2C-safe.

Quick Params (current defaults)

Param	Value	Description
BASE_SPEED	65	Forward/Backward baseline speed
SMOOTH_TURN_SPEED	50	Continuous turn speed (reduced for I2C stability)
SHARP_TURN_SPEED	18	Sharp turn speed (short pulse on micro:bit)

Gesture → Command Mapping

Gesture (Emoji)	Hand	Command	Behavior	Notes
 Open Palm	Either	CMD:FWD	Move forward (continuous).	Resend every ~0.5 s to keep BLE alive.
 Thumb Up	Either	CMD:BACK	Move backward (continuous).	Same resend policy as FWD.
 Fist	Either	CMD:STOP	Immediate stop.	Bypasses rate limit when ready.
 One Finger (Middle)	Left	CMD:SMOOTH_LEFT	Continuous pivot/arc turn left.	I2C-safe soft-start ramp enabled.
 One Finger (Middle)	Right	CMD:SMOOTH_RIGHT	Continuous pivot/arc turn right.	I2C-safe soft-start ramp enabled.
 Peace	Left	CMD:SHARP_LEFT	Short, sharp left pulse.	One-shot; on-board pulse + brief brake; requires release.
 Peace	Right	CMD:SHARP_RIGHT	Short, sharp right pulse.	One-shot; on-board pulse + brief brake; requires release.

BLE / UI Info

UI fields: BLE status, MAC address, Gesture, Command, Sharp state, FPS.

Sharp is edge-triggered: hold 🖐️ does not repeat; release and re-pose to trigger again.

Tip: Tune sharpness on-board via SHARP_PULSE_MS and BRAKE_MS; tune responsiveness on PC via stabilizer params.